

DIVISIONS — where a piece of work belongs

The divisions are the table below, and `node engine/orient.js` derives the live list from `.claude/agents/` — **do not type a count here**. This line read *"Four divisions"* from 2026-08-04, when WEB was added, until 2026-08-23, and CLAUDE.md said "four" in three places over the same period. The count is cosmetic; what it cost was not — the living-docs rule named four ledgers, so a WEB change carried no documented obligation to record itself anywhere.

The cut is on the **invalidation graph**, not on subject matter, because what makes a handoff expensive here is not how much there is to say — it is that any change might invalidate anything else, so the whole project has to be described every time.

Do not cut by Pokémon domain ("a weather division", "a Protect division"). That runs across the graph below and makes every change touch every division, which is the situation this replaces.

The graph

Reg M-C, from 2026-09-24. Reg M-B is retired. The target is Reg M-C with open team sheets, and every model except MEDICHAM is rebuilt from scratch under `solver/` (`solver/PLAN.md`).

MEDICHAM (a frozen release) → SOLVER → the ladder
(engine, verified) (CHOMP · XATU · MAG/DODUO · MILTANK · SLOWKING · PORYGON2 · GARY/HYPNO · MEW/MACHAMP · ROTOM)

the store (OPS) feeds SOLVER's store-only models directly: the human dataset, MAG, DODUO, XATU, GURU
MEASURE / SPRT / provenance / status sit BESIDE all of it and invalidate nobody

The engine is the foundation; the search is the point. SOLVER plays a frozen MEDICHAM release, so ENGINE can keep fixing the simulator while SOLVER measures.

Reg M-B, until 2026-09-20 — retired, kept as it was drawn:

MEDICHAM → board.js → MAG weights → MILTANK baselines → live
(engine) (features) (the refit) (every H2H result)

MEW / SPRT / provenance / status sit BESIDE all of it and invalidate nobody

It is one-way. That is the only reason dividing is worth doing — if ENGINE ever came to depend on a SOLVER result, one big document would beat several small ones and this file should be deleted.

The divisions

Division	Owns	Its one number	Ledger
ENGINE	medicham2-browser.js , abra-tags , tests/test-mechanics.js , tests/walk_tags.js , tests/test-engine-diff.js	mechanics live (must never go down)	ENGINE.md
MEASURE	mew.js , sprt.js , provenance.js , status.js , backtest_winrate.js , the noise floor, the stamps	leaf calibration	MEASURE.md
SOLVER	solver/ — every model in solver/PLAN.md §2 except MEDICHAM: preview (CHOMP), belief (XATU), candidate narrowing	the settled ladder rating ± SD and the	SOLVER.md

Division	Owns	Its one number	Ledger
	(MAG, DODUO), the turn search (MILTANK, SLOWKING), value (PORYGON2), habits and the exploit dial (GARY, HYPNO), self-play (MEW, MACHAMP), the live client (ROTOM)	per-series residual; offline, SPRT at equal wall-clock	
OPS	Showdown replays, OTS, ingest, the stores and the frozen team pools	store usable %, battles recorded	OPS.md
WEB	web/ — ABRA WORLD and every room in it	every rendered figure traces to an artifact	WEB.md

(2026-09-24. *SEARCH* was renamed *SOLVER*: Will — search is the core of the plan and is not going away, so the division takes on the nets and the live client. *OPS* keeps ingest and the store, and hands the live bot to *SOLVER*, where *ROTOM* replaces `mag_bot.js`.)

MAG is not a division — it is the seam. It consumes *ENGINE* and feeds *SEARCH*, and its refit is the expensive event on the one expensive edge. The refit therefore belongs to *MEASURE*, whose whole job is knowing when a number stopped being true.

(Retired with Reg M-B, left as written. *MAG* is rebuilt in `solver/mag/` as a store-only model with no simulator in its path, so it is no longer a seam: it is *SOLVER*'s. *MEASURE* still judges whether its numbers are true.)

Routing a bug: one question

Which artifact does fixing this invalidate?

- The damage table → **ENGINE**
- A measurement claim, a stamp, a corpus → **MEASURE**
- What gets clicked, but not what is true → **SOLVER**
- Nothing → **OPS**

A bug that cannot be routed does not get held. It gets a division or it gets closed.

The two rules that make the division real

1. *SOLVER* plays a frozen, named engine release — never HEAD

ENGINE batches fixes and cuts a release. Between releases, *SOLVER*'s baselines are frozen and valid, and *ENGINE* can land twenty mechanics fixes without invalidating a running H2H. On Reg M-C a figure that played on a release before `eea5becc54eb` (the release the M-C gate opened on) is PRE-GATE and is withheld, not captioned.

(Until 2026-09-24 this rule was *SEARCH*'s, and the paragraphs below describe the Reg M-B stack: the refit and the seven restamps are retired with it.)

This is not theoretical. `node engine/status.js` currently prints every R4 run as `PRE-CHANGE`: the engine source moved after the games were played, so the headline result of 2026-08-04 already describes a build that no longer exists.

It also fixes the schedule. *ENGINE*'s work — tag probes, differential tests — is single-process and cheap. *SOLVER* and *MEASURE* eat the process budget. Under a release boundary those genuinely run side by side; without one they collide.

2. If you trip over another division's bug, you file it — you do not fix it

Patching a mechanics bug in the middle of an H2H silently invalidates your own run, and the run will not tell you. This is Lesson 1 wearing a different hat: the result still prints.

What this costs

A division boundary is a new place for a **silent default** to live — a division working from a stale artifact looks exactly like a division working. Two mitigations, both already built:

- `engine/provenance.js` checks artifact-against-artifact and is the authority on staleness.
- `engine/status.js` checks the edge provenance structurally cannot see — engine SOURCE against fitted weights — and prints `REFIT OWED` . It is mtime-based and says so.

Neither can catch an artifact that records a corpus it did not use. Only re-running the generator can, and that is the generator's job.

One agent per division

`.claude/agents/{engine,measure,solver,ops,web}.md` . Each loads CLAUDE.md plus its own ledger and nothing else, so it cannot reason wrongly about a part of the project it was never shown.

WEB was added 2026-08-04, and the reason is worth recording because it tests whether the cut above is real. `web/` had no owner: ENGINE, MEASURE, SEARCH and OPS are all cuts on the model's invalidation graph, and a website is not on that graph. So site work kept falling back to whoever was holding it. WEB is the **leaf** — everything flows into it and nothing flows out — which is exactly why it can be given hands on its own files and none anywhere else, and why its one restriction is not about tools but about authority: it renders numbers and never authors one.

The point is not that an agent is an expert — it is a fresh context holding the right slice. What the split actually buys is that the **may-not column becomes a tool restriction instead of a sentence somebody might ignore:**

Agent	Hands	The restriction that matters
engine	full	may not touch <code>solver/</code> , <code>board.js</code> / <code>magnemite.js</code> / <code>engine-data.js</code> , may not run a fit or self-play
measure	full	must ask before starting a refit — it is expensive and Will may be at the keyboard
solver	full, never <code>engine/</code>	prepares wide runs and ladder series and hands over the command; never launches a ladder series — that spends a real rating and is Will's call
ops	read-only	no Bash, no Write, no Edit — a mistake here forfeits a real game
web	full, inside <code>web/</code> only	may not author a number — every figure traces to an artifact or renders as NOT MEASURED

This is not parallelism. Six processes is the cap and RAM is the real ceiling. ENGINE's work is single-process and genuinely runs alongside a long SOLVER job; two solver agents do not. The win is clean scope and a small context, not throughput.

The handoff

There is no longer a handoff document to write.

```
node engine/status.js
```

That output is the handoff. `--write` also stamps it into the division ledgers (from the main checkout — never from a worktree). The rules live in `CLAUDE.md` and do not change; the lessons live in `docs/LESSONS.md` and are written once.

The `HANDOFF-*.md` files are history now, not state. Most of them moved to `docs/archive/` on 2026-08-05, each carrying a header saying what it claimed, what replaced it, and which of its figures are retracted; `docs/archive/INDEX.md` is generated from those headers by `build/build_archive_index.js`. Read one for narrative if you want it, but do not take a number out of it — the 2026-08-04 handoff says "172 tags, 118 unprobed" and the artifact says 176 and 123. Nobody mistyped anything. Prose cannot track a corpus.

Archiving does not launder a document. `tests/test-docs-current.js` scans `docs/archive/` exactly as it scans a live document unless the file declares a named replacement, and the sixteen archived on 2026-08-05 deliberately declare none — so a retracted figure inside one is still caught.